

Self-Supervised Pretraining for Autonomous Driving: A Survey

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Abstract

Autonomous driving models need representations that support perception, prediction, and planning, but dense annotations are costly to obtain. This survey reviews self-supervised pretraining on autonomous driving data through an objective-first taxonomy covering masked reconstruction, reconstruction-based tokenization, contrastive learning, rendering-based objectives, future prediction, trajectory-map pretraining, and joint-embedding predictive architecture (JEPA)-style latent prediction. Rather than forming a single leaderboard, the survey emphasizes how each pretext task preserves spatial, geometric, temporal, multimodal, or behavioral structure for downstream tasks such as detection, segmentation, occupancy forecasting, motion forecasting, and planning-oriented evaluation. The literature shows a shift from generic feature learning toward driving-specific objectives, while fair comparison remains limited by differences in datasets, backbones, supervision budgets, pretraining targets, and fine-tuning protocols.

1 Introduction

Autonomous driving systems depend on perception and prediction models that remain reliable across diverse road layouts, traffic densities, weather conditions, lighting conditions, and geographic domains. While large-scale driving logs are relatively easy to collect, dense annotations are expensive to obtain. Self-supervised pretraining therefore offers a natural way to exploit the large amount of unlabeled data already collected by vehicle fleets.

However, the key question is not simply whether unlabeled data improves downstream performance. For autonomous driving, pretraining is especially challenging because the target tasks are not generic image-recognition

problems. Driving models must capture spatial structure, 3D geometry, temporal dynamics, and multimodal interactions. As a result, the effectiveness of self-supervised pretraining depends not only on feature transferability, but also on whether the learned representation is aligned with the structure required by downstream driving tasks.

This survey therefore organizes the literature according to pretraining objective rather than treating each paper as an isolated method summary. We categorize existing approaches into masked reconstruction, reconstruction-based tokenization, contrastive learning, rendering-based pretraining, future prediction, trajectory-map pretraining, and joint-embedding predictive architecture (JEPA)-style latent prediction. This objective-based taxonomy allows us to examine how different pretraining targets and representation spaces relate to downstream tasks.

2 Background

2.1 Self-Supervised Pretraining

Self-supervised learning (SSL) pretrains a model with supervisory signals derived from the data itself rather than from manual labels. In many representation-learning settings, a model backbone or encoder is first optimized on unlabeled samples through a pretext task, and the learned representation is then transferred to downstream tasks by fine-tuning. In visual representation learning, common SSL families include contrastive learning, masked reconstruction, and teacher-student or latent-prediction objectives [1–4]. The common goal is not to solve the pretext task for its own sake, but to learn features that remain useful when labeled data is limited.

For autonomous driving, this idea is especially attractive because raw driving logs can usually be collected at a much larger scale than dense manual labels. A vehicle

equipped with cameras, LiDAR, radar, Global Navigation Satellite System (GNSS), and inertial sensors can record synchronized multimodal streams during routine road tests, fleet operation, simulation, or data-collection drives. Once the sensor platform and storage pipeline are in place, the unlabeled dataset can be expanded largely by driving more routes and retaining the recorded observations. In contrast, converting those logs into supervised training data requires human or semi-automatic annotation of 3D geometry, object identities, semantic regions, tracks, occupancy states, and sometimes future behavior, all of which are costly and error-prone.

2.2 Driving Data and Sensor Modalities

Autonomous driving data differs from generic visual datasets because it is multimodal, geometric, temporal, and ego-centric. Cameras provide dense appearance cues but do not directly measure metric depth in the way range sensors do. LiDAR provides accurate 3D geometry but is sparse, range-dependent, and sensitive to sensor configuration and surface properties. Radar is useful for velocity estimation and robustness under adverse conditions, but its spatial resolution is limited. GNSS and inertial sensors provide ego-motion context, while synchronized multi-camera and multi-sensor logs introduce calibration and temporal-alignment constraints.

These modality-specific properties determine what kinds of self-supervised signals are natural in driving data. A useful pretraining task should respect the geometry, sparsity, temporal continuity, and calibration structure of the recorded sensors rather than treating each frame as an independent natural image. This motivates driving-specific SSL objectives that derive supervision from sensor-specific cues while remaining useful for downstream perception, prediction, and planning tasks.

2.3 Common Datasets and Benchmark Protocols

Dataset choice is part of the experimental protocol, not a neutral implementation detail. The driving SSL literature commonly uses KITTI [5], SemanticKITTI [6], nuScenes [7], Waymo Open Dataset [8], ONCE [9], Argoverse 2 [10], SemanticPOSS [11], as well as simulation platforms such as CARLA [12]. These resources differ in

sensor layout, geographic diversity, annotation type, and task definition. For example, LiDAR semantic segmentation on SemanticKITTI or SemanticPOSS is not directly comparable to 3D detection on nuScenes, large-scale unlabeled pretraining on ONCE, motion forecasting on Argoverse 2, or future-scene generation in CARLA. General 3D self-supervised baselines may also report results on non-driving 3D benchmarks such as ScanNet, S3DIS, SUN RGB-D, and ShapeNetPart. In addition, synthetic urban-scene resources such as SYNTHIA are sometimes used to study simulation-to-real transfer. Therefore, when comparing pretraining methods, dataset citations should be read together with backbone, labeled-data fraction, downstream head, and fine-tuning schedule.

3 Method Taxonomy and Comparison

This taxonomy is organized by pretraining objective rather than by treating each paper as a separate category. Conventional self-supervised pretraining methods usually learn an encoder or representation that is directly transferred to downstream tasks. Recent driving world models are harder to place because they often combine tokenizers, latent predictors, diffusion or autoregressive modules, and planning-oriented components. This survey therefore treats them as boundary cases and discusses their components when those components correspond to a specific pretraining objective. For example, BEVWorld’s multimodal tokenizer is relevant to reconstruction-based tokenization, whereas its latent BEV sequence diffusion is relevant to future prediction [13]. Accordingly, Table 1 focuses on methods closer to conventional self-supervised pretraining, where the learned representation is directly evaluated through downstream transfer.

3.1 Masked Reconstruction-Based Pretraining

Masked reconstruction adapts the masked autoencoding idea to sparse and structured driving observations. The model receives a partially hidden scene and learns to recover a target derived from the missing region. In autonomous driving, this target is rarely a generic image

Table 1: Comparison of representative self-supervised pretraining methods.

Method	Input	Dataset(s)	Representation	Objective	Downstream evidence
Voxel-MAE [14]	LiDAR	nuScenes	Voxel space	Masked voxel reconstruction	Transfer to 3D object detection
Occupancy-MAE [15]	LiDAR	ONCE, KITTI, nuScenes, Waymo	Voxel space	Masked binary voxel occupancy prediction	Transfer to 3D object detection, semantic segmentation, multi-object tracking, and domain adaptation
BEV-MAE [16]	LiDAR	nuScenes, Waymo	BEV space	BEV-guided masked density prediction	Transfer to 3D object detection
GeoMAE [17]	LiDAR	nuScenes, Waymo	Voxel space	Masked voxel geometric target prediction	Transfer to 3D object detection, semantic segmentation, and multi-object tracking
TARL [18]	LiDAR	SemanticKITTI, KITTI, nuScenes	Point space	Temporal segment-level contrastive learning	Transfer to 3D object detection, semantic segmentation, and panoptic segmentation
BEV-Contrast [19]	LiDAR	SemanticKITTI, SemanticPOSS, KITTI, nuScenes	BEV space	Temporal BEV-level contrastive learning	Transfer to 3D object detection and semantic segmentation
UniPAD [20]	Camera, LiDAR, and multimodal	nuScenes	Voxel space	Rendering-based reconstruction	Transfer to 3D object detection and semantic segmentation
ViDAR [21]	Camera input; LiDAR supervision	nuScenes	BEV space	Visual point cloud forecasting	Transfer to 3D object detection, 3D semantic occupancy prediction, multi-object tracking, map segmentation, motion forecasting, 4D binary occupancy forecasting, and planning
Forecast-MAE [22]	Trajectories, road network	Argoverse 2	Trajectory-map token space	Masked trajectory-map reconstruction	Transfer to motion forecasting
Plan-MAE [23]	Trajectories, road network, and route	Self-collected driving data	Trajectory-map-route token space	Masked trajectory-map-route reconstruction and local sub-planning	Transfer to motion forecasting, planning
AD-L-JEPA [24]	LiDAR	ONCE, Waymo, KITTI	BEV space	Masked spatial BEV embedding prediction	Transfer to 3D object detection
AD-LiST-JEPA [25]	LiDAR	Waymo	BEV space	Masked spatiotemporal BEV embedding prediction	Transfer to 4D binary occupancy completion and forecasting

patch: it can be voxel occupancy, BEV-level point density, or local geometric structure. The value of the pretext task therefore depends on whether the reconstruction target preserves the spatial information that downstream driving models need.

Voxel-MAE reconstructs masked voxelized LiDAR structure and uses this objective to initialize LiDAR perception models [14]. Occupancy-MAE also operates in voxel space, but frames the target as masked binary occupancy prediction over large-scale outdoor LiDAR point clouds [15]. BEV-MAE moves the design closer to detector geometry by using BEV-guided masking and point-density prediction, making the pretraining target more explicitly tied to LiDAR object detection [16]. GeoMAE asks which geometric signals are useful by predicting masked geometric targets such as local centroids, normals, curvature, and binary occupancy [17].

The main comparison within this family is the reconstruction space. Voxel and occupancy targets preserve metric 3D structure, but they inherit sparsity and resolution choices from LiDAR voxelization. BEV-oriented

targets are better aligned with many downstream detection and planning representations, but may make height and fine-grained 3D geometry less explicit. Geometry-specific targets such as GeoMAE’s local geometric quantities reduce the risk of learning only sensor density patterns, but they still rely on hand-designed target choices. Across this family, reconstruction supplies abundant structured self-supervision, while the open question is how much of the reconstructed signal is actually useful for perception, forecasting, or planning.

3.2 Reconstruction-Based Tokenization

Reconstruction-based tokenization also uses decoder-side reconstruction, but its role differs from masked autoencoding. Instead of hiding part of the input and training an encoder to infer missing regions from visible context, these methods compress full observations into compact latent tokens and train a decoder to reconstruct the original image or occupancy representation. The learned tokens then become the state space for a later autoregressive, diffusion,

or forecasting model, so the tokenizer is a component of a larger world-model pipeline rather than a direct downstream encoder initialization.

GAIA-1 trains a discrete image tokenizer before the world model. Each video frame is mapped into image tokens by a discrete autoencoder, whose training combines image reconstruction, quantization, and DINO-distillation losses [26]. OccWorld follows the same broad tokenization logic in a 3D semantic occupancy space: it trains a VQ-VAE-style scene tokenizer to encode occupancy into discrete scene tokens and reconstruct occupancy through the decoder [27]. BEVWorld converts heterogeneous sensor inputs into scene-level BEV latents; its multimodal tokenizer reconstructs LiDAR and surround-view images from those BEV tokens before the latent sequence model forecasts future scenes [13]. Copilot4D is similar at the tokenizer stage: its VQ-VAE-like LiDAR tokenizer maps point-cloud observations into BEV discrete latents and reconstructs observations through differentiable depth rendering [28]. RenderWorld applies this idea to generated 3D semantic occupancy labels with an Air Mask Variational Autoencoder (AM-VAE), which separately encodes air and non-air voxels [29].

The distinction is important for comparison. These tokenizers may reconstruct images or occupancy grids, but the representation being evaluated is usually the token space used by a later world model, not a backbone transferred directly to detection or segmentation. Their reconstruction losses therefore serve compression and discretization: a useful tokenizer should preserve enough semantic, geometric, and dynamic information for sequence modeling, while avoiding a token vocabulary that overfits low-level reconstruction detail and becomes difficult to forecast.

3.3 Contrastive Learning-Based Pretraining

Contrastive learning constructs supervision by defining which observations should have similar features and which should be separated. This is natural for driving logs because the ego vehicle observes related scene content over time, but the definition of a positive pair is delicate. Traffic scenes contain repeated structures, independently moving agents, and changing visibility, so positive and negative sampling can either reveal temporal consistency or introduce misleading invariances.

TARL uses temporal consistency in LiDAR sequences

to learn representations for autonomous-driving semantic perception [18]. Its contrastive unit is closer to an object or segment observed across time than to an entire scan. BEVContrast instead formulates contrastive learning in BEV space, contrasting 2D BEV cell representations across scans from the same scene [19]. PointContrast is useful as background, as it demonstrates point-level 3D contrastive pretraining, but it is not as driving-specific as TARL and BEVContrast [30].

The contrast between TARL and BEVContrast is mainly granularity. TARL emphasizes temporally associated object- or segment-level structure, which is well matched to semantic perception but depends on temporal association and the quality of extracted segments. BEVContrast avoids expensive object-level preprocessing by using BEV cells, which makes the contrastive unit closer to downstream BEV perception. However, BEV-level positives still depend on accurate spatial alignment, and both approaches must handle the fact that the same location may change over time while different locations may contain visually similar road structure.

3.4 Rendering-Based Pretraining

Rendering-based pretraining supervises a learned 3D, voxel, or BEV representation by asking it to explain observed sensor measurements. Instead of reconstructing only a masked local region, the model must connect internal spatial features to camera images, LiDAR points, or both. This objective is especially relevant for multimodal autonomous driving because cameras provide dense appearance and LiDAR provides metric geometry.

UniPAD is the clearest example of this category: it proposes a universal autonomous-driving pretraining paradigm based on volumetric differentiable rendering, enabling the learned representation to transfer across camera, LiDAR, and multimodal frameworks [20]. PreWorld and RenderWorld are related occupancy-centric world-modeling examples. PreWorld uses volume rendering as a pretraining signal to exploit 2D supervision for 3D semantic occupancy world modeling, while RenderWorld uses a Gaussian-based Img2Occ module to generate self-supervised 3D semantic occupancy labels before world-model training [29, 31].

BEVWorld and Copilot4D are component-level boundary cases: their tokenizer training also uses rendering-

related reconstruction signals, but these losses mainly serve compact BEV or LiDAR tokenization for later world modeling rather than direct encoder pretraining [13, 28]. We therefore discuss their mechanisms under reconstruction-based tokenization and treat them here only as examples of how rendering objectives can appear inside larger generative driving pipelines. Overall, the main advantage of rendering objectives is that they tie compact spatial representations to observable sensor data without requiring dense manual 3D labels.

3.5 Future Prediction-Based Pretraining

Future prediction changes the pretext task from reconstructing the current observation to modeling how the driving scene evolves. This family is conceptually closer to autonomous driving as a sequential decision problem, because downstream behavior depends on future geometry, occupancy, agent motion, ego-vehicle actions, and planning outcomes. The tradeoff is that future-prediction methods often introduce pretrained tokenizers, diffusion or autoregressive modules, planner interfaces, and task-specific evaluation protocols, making them harder to compare against static detector pretraining.

ViDAR predicts future point clouds from historical visual inputs, using future LiDAR observations as a driving-specific supervision signal for visual autonomous driving [21]. UnO follows a related geometry-driven direction, but takes past LiDAR sweeps as input and formulates the future state as a continuous 4D binary occupancy field learned from unlabeled LiDAR [32]. DriveWorld is another representation-oriented pretraining method: it learns from multi-camera driving videos using a world-model framework for 4D scene understanding, where a memory state-space design models temporal dynamics and spatial context, and pretraining is driven mainly by current and future 3D occupancy prediction, action prediction, and latent dynamics modeling [33].

A second group consists of generative or tokenized driving world models, where future prediction is embedded in a larger pipeline that usually maps sensor observations or occupancy states into a compact latent space. OccWorld predicts future scene and ego-trajectory tokens in a discrete occupancy-token space [27]. BEVWorld forecasts scene evolution through action-conditioned BEV latent diffusion [13]. Copilot4D predicts future BEV to-

kens with discrete diffusion [28]. RenderWorld forecasts future semantic occupancy for planning based on self-supervised 3D occupancy labels and AM-VAE latents [29]. Drive-OccWorld forecasts future semantic occupancy and flow from historical BEV features conditioned on actions, and connects the generated occupancy to trajectory selection [34]. GAIA-1 performs autoregressive image-token prediction, while DriveDreamer and Drive-WM shift the objective toward controllable driving-video generation and planning-oriented visual forecasting [26, 35, 36].

The main comparison within this family is the definition of the future state. ViDAR uses future point-cloud geometry, UnO predicts continuous 4D binary occupancy fields, DriveWorld emphasizes transferable 4D visual representations, OccWorld, RenderWorld, and Drive-OccWorld forecast occupancy-centric states, BEVWorld and Copilot4D operate in BEV latent or token spaces, and GAIA-1, DriveDreamer, and Drive-WM focus more on visual generation or simulation. Within the occupancy-centric group, Drive-OccWorld places more emphasis on action-conditioned forecasting and trajectory selection, whereas OccWorld and RenderWorld are more directly framed around occupancy-token or occupancy-label forecasting pipelines. These choices make the methods temporally aligned with driving, but they also make evaluation less standardized: the evidence spans perception, forecasting, occupancy prediction, simulation quality, and planning-oriented metrics rather than a single downstream transfer protocol.

3.6 Trajectory-Map Pretraining

Trajectory-map pretraining starts from the observation that driving is not only perception. A representation useful for autonomous vehicles should capture interactions, map constraints, route intent, and feasible ego behavior. These methods therefore operate on structured driving data such as trajectories, lanes, road networks, and route information rather than only on raw sensor measurements.

Forecast-MAE adapts masked autoencoding to motion forecasting by masking agent trajectories and lane segments, encouraging the model to learn dependencies among agents, history, future motion, and road structure [22]. Plan-MAE extends this direction toward integrated prediction and planning by reconstructing masked road networks, agent trajectories, and navigation routes,

and by adding a local sub-planning task intended to encode vehicle dynamics and safety constraints [23].

The comparison is mainly one of scope. Forecast-MAE is aligned with motion forecasting and representation learning over trajectory-map tokens, whereas Plan-MAE expands the objective with route-conditioned reconstruction and a local sub-planning task, making it closer to the information required by behavior planning. Although both methods use MAE-style reconstruction, their reconstruction space is trajectory-map structure rather than raw sensor, voxel, or BEV perception space. They are therefore better viewed as trajectory-map pretraining methods rather than direct replacements for sensor-oriented SSL.

3.7 JEPA-style Latent Prediction

JEPA-style methods avoid reconstructing raw observations and instead predict latent embeddings. This changes the objective from pixel-, point-, voxel-, or occupancy-level recovery to representation-level consistency. The motivation is that a compact latent target may preserve semantic and geometric structure while avoiding unnecessary reconstruction of sensor noise, texture, or sparsity artifacts.

AD-L-JEPA applies joint-embedding predictive architecture ideas to automotive LiDAR object detection by predicting BEV embeddings and using variance regularization to avoid representation collapse [24]. AD-LiST-JEPA extends the same broad direction to a spatiotemporal LiDAR world-model setting and evaluates the learned representation through binary occupancy completion and forecasting [25]. Compared with masked reconstruction, these objectives place less emphasis on recovering the input itself and more emphasis on predicting an abstract target representation.

The promise of JEPA-style pretraining is semantic and geometric abstraction. If the latent target keeps the geometry and dynamics needed by downstream tasks, the model can avoid spending capacity on low-level reconstruction details. The risk is the opposite: compact embeddings may discard rare objects, occluded regions, or uncertainty unless the pretraining target and downstream evaluations explicitly preserve those factors. For autonomous driving, JEPA-style objectives are therefore best judged by whether their latent spaces transfer beyond the task used to validate them.

4 Discussion

4.1 From Objective Families to Task Alignment

The objective-first taxonomy should be read as an alignment question rather than a ranking of objective families. A driving representation is useful only insofar as it preserves the structure required by downstream tasks: metric geometry for 3D detection, free-space and occlusion cues for occupancy reasoning, spatial abstraction for BEV perception, temporal continuity for forecasting, and behavior-relevant constraints for planning. Self-supervised pretraining is therefore not just about whether an objective is reconstruction-based, contrastive, predictive, rendering-based, trajectory-map-based, or latent-predictive. The more important question is which part of the driving problem the objective asks the representation to model.

This perspective explains why the objective families are complementary rather than strictly ordered. Masked voxel, occupancy, and BEV reconstruction methods provide dense geometric supervision and are naturally aligned with perception transfer [14–16]. Contrastive methods emphasize invariances across time or views, which can help when the downstream model needs stable scene identity rather than exact sensor recovery [18, 19]. Rendering-based objectives connect compact spatial representations to observable camera or LiDAR measurements, making them useful for multimodal grounding [20]. Future-prediction and world-modeling methods shift the target from present observations to scene evolution, occupancy forecasting, or planning-oriented behavior [21, 27, 34]. Trajectory-map pretraining moves from raw perception toward structured prediction by encoding agent trajectories, map elements, routes, and, in Plan-MAE, local planning structure [22, 23]. JEPA-style methods offer another direction by predicting latent BEV embeddings instead of reconstructing raw observations [24, 25]. The overall trend is a move from local sensor recovery toward spatially grounded, temporally predictive, and decision-relevant representations.

4.2 Why Current Evidence Does Not Support a Global Ranking

Current cross-paper comparisons should be read as qualified evidence rather than direct rankings. Reported gains

are tied to the baseline used in each paper: improvement over random initialization is not equivalent to improvement over ImageNet pretraining, supervised driving pretraining, an earlier LiDAR SSL method, or a larger multimodal model. The downstream tasks also measure different transfer properties. A detector benchmark mainly tests whether the pretrained encoder preserves geometry useful for object localization, while segmentation, occupancy prediction, motion forecasting, and planning-oriented evaluation require different kinds of spatial, temporal, or behavioral abstraction [14, 15, 21, 22].

Implementation choices further weaken direct ranking. Papers differ in unlabeled data scale, dataset domain, sensor setup, backbone capacity, labeled-data fraction, downstream head design, and fine-tuning schedule. Even methods that appear to share an objective family may differ in what is actually reconstructed or predicted: raw voxels, occupancy, BEV density, rendered observations, future point clouds, occupancy or scene tokens, trajectory-map tokens, or latent embeddings [14–16, 20–22, 24, 27, 32]. These targets are not interchangeable learning signals, so a higher reported downstream number does not necessarily imply a better pretraining objective in general.

The issue becomes sharper for world-model and trajectory-map or planning-oriented systems. In these methods, representation learning is often entangled with a tokenizer, generative predictor, action-conditioning, planner, or route-aware objective [13, 23, 26, 28, 34]. Such designs are reasonable for building complete driving systems, but they make it difficult to isolate whether the reported improvement comes from the pretrained representation itself, the future predictor, the generation pipeline, the planner design, or private data scale. For this reason, the safest interpretation is controlled evidence within each paper’s protocol, not a global ranking across objective families.

4.3 Toward Alignment-Oriented Benchmarks

A future benchmark for driving SSL should evaluate objective–task alignment under controlled conditions. It should match backbone families, unlabeled pretraining splits, labeled-data fractions, downstream heads, and fine-tuning schedules, and evaluate the same pretrained representation across detection, segmentation, BEV perception,

occupancy prediction, motion forecasting, robustness, and planning-oriented tasks when applicable.

Such a benchmark should include diagnostics beyond a single leaderboard score. Useful tests would measure geometry preservation after projection, semantic abstraction beyond raw sensor density, temporal consistency, uncertainty under occlusion, cross-dataset transfer, and sensitivity to rare or safety-critical objects. These diagnostics would clarify whether a method learns a representation aligned with driving structure or only improves a narrow downstream head under one training protocol.

For rendering-based, world-model, and trajectory-map or planning-oriented pretraining, benchmark design should also separate representation transfer from generation quality, tokenizer reconstruction, predictor capacity, planner design, action conditioning, and data scale. This separation would make it possible to compare masked reconstruction, contrastive learning, rendering-based pretraining, future prediction, trajectory-map pretraining, and JEPA-style latent prediction under a common evaluation contract. The goal is not to declare a universal best objective, but to identify which objective–representation pairs align with which downstream driving requirements.

5 Conclusion

Self-supervised pretraining provides a practical way to exploit the large-scale unlabeled driving data collected by autonomous-driving platforms. The surveyed methods illustrate a shift from generic image-level SSL ideas toward driving-specific objectives that preserve spatial, geometric, multimodal, and temporal structure. Masked reconstruction methods emphasize LiDAR-derived geometric structure in BEV, voxel, or occupancy spaces; contrastive methods define correspondences across points, objects, segments, time, or BEV cells; rendering and multimodal reconstruction connect camera appearance with 3D geometry; and future-prediction, trajectory-map-based, and JEPA-style latent objectives shift the target beyond current-observation reconstruction toward scene dynamics and driving-relevant states.

A central remaining challenge is alignment between the pretext objective and downstream driving requirements. Raw reconstruction gives dense supervision but may spend capacity on sensor-level detail, while latent prediction and

distillation can encourage abstraction but depend on the quality of the target representation. Future work should therefore pair stronger multimodal and temporal pretraining with more controlled evaluation protocols across detection, segmentation, occupancy prediction, forecasting, robustness, cross-dataset transfer, and planning-oriented tasks. A useful long-term direction is a unified driving representation that combines BEV or voxel geometry, occupancy state, and temporal dynamics while remaining efficient enough for practical autonomous driving systems.

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